A Geostatistical Approach to Peatland Soil Moisture Analysis in Sumatra Using Ordinary and Universal Kriging

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***Abstract*— Soil moisture is a critical indicator of peatland hydrology and fire susceptibility, yet monitoring in Indonesia relies on sparse point observations. This study applies ordinary kriging (OK) and universal kriging (UK) to generate continuous soil moisture maps for the peatlands of Riau, Jambi, and South Sumatra using the 2023 Peatland Restoration Information and Monitoring System (PRIMS) dataset. Experimental semivariograms were fitted with spherical, exponential, and Gaussian models, and predictive performance was assessed using spatial-block cross-validation to reduce optimistic estimates caused by spatially random validation. The comparison between OK and UK showed province-dependent performance, with UK providing lower errors in Jambi, while OK achieved lower errors in Riau and South Sumatra. These differences indicate that the benefit of incorporating deterministic drift depends on the spatial characteristics of each region. Predictability varied with spatial structure, with Riau exhibiting weak spatial autocorrelation associated with a near-nugget variogram, while Jambi showed stronger spatial predictability. The results highlight the importance of variogram characterization and spatially appropriate validation for reliable peatland soil moisture mapping, while the choice between OK and UK should be considered according to regional spatial behavior. The resulting soil moisture maps provide spatial information that can support future peatland monitoring and fire-risk assessment frameworks.**

Keywords—*soil moisture, peatland, semivariogram, kriging, Sumatra*

I. Introduction

Peatlands are vital ecosystems that provide hydrological regulation, long-term carbon storage, and climate stabilization, functioning as natural buffers that control water retention, reduce flood peaks, and sequester substantial amounts of organic carbon over millennial timescales [1], [2], [3], [4], [5], [6]. However, these functions are highly sensitive to soil moisture dynamics, as prolonged drying lowers the water table, increases peat aeration, and accelerates organic matter oxidation, thereby sharply elevating flammability and fire susceptibility [7], [8]. Soil moisture therefore serves as a direct and physically meaningful indicator of peatland fire risk, whose degradation can lead to severe environmental and economic losses [9], [10].

Sumatra's peatlands, concentrated in Riau, Jambi, and South Sumatra, have repeatedly been among the areas most affected by recurrent peatland fires in Indonesia, particularly during prolonged dry seasons [11]. The risk is strongly modulated by climate: during El Niño years, extended drought lowers peatland water tables and drives large-scale fire events, as observed in major fire seasons across the region [12]. The year 2023 coincided with a renewed El Niño phase [13], making the characterization of dry-season soil moisture during that period particularly relevant for understanding fire-prone conditions. These three provinces are therefore prioritized in this study.

In Indonesia, soil moisture monitoring through the Peatland Restoration Information and Monitoring System (PRIMS) of the Peatland and Mangrove Restoration Agency (BRGM) provides essential in situ observations, yet these measurements are collected at discrete monitoring points, resulting in limited spatial continuity at the landscape scale. This spatial sparsity constrains the ability to identify critical drying zones, assess the connectivity of fire-prone areas, and support proactive fire prevention and restoration planning. There is therefore a strong need for spatially explicit modeling frameworks that can translate point-based observations into continuous regional assessments, strengthening peatland protection and the effectiveness of fire-risk mitigation.

Geostatistical methods, particularly semivariogram analysis and kriging, are widely applied to map soil properties from scattered observations into continuous surfaces [14], [15], [16]. As a leading GIS (Geographic Information Systems)-based digital mapping technique [16], kriging models spatial autocorrelation to provide more reliable predictions than deterministic interpolation and quantifies prediction uncertainty, supporting robust environmental decision-making [17], [18]. Ordinary kriging assumes a constant mean, whereas universal kriging incorporates a deterministic spatial trend for non-stationary data. Because soil moisture over the large geographic extent of Sumatra may vary systematically with regional gradients in rainfall, elevation, and drainage, a large-scale spatial trend can

plausibly be expected; comparing ordinary and universal kriging therefore allows this assumption to be tested rather than assumed.

To leverage these capabilities, this study applies ordinary kriging (OK) and universal kriging (UK) to map peatland soil moisture in Riau, Jambi, and South Sumatra using the 2023 PRIMS dataset. Experimental semivariograms are first constructed and fitted with alternative theoretical models to characterize spatial dependence and select the most appropriate variogram model. The predictive performance of OK and UK is then evaluated through spatial-block cross-validation using RMSE, MAE, and MSDR to assess both prediction accuracy and uncertainty calibration, before generating continuous soil moisture maps for each province. This integrated workflow provides a systematic assessment of both spatial dependence and interpolation performance to support peatland monitoring and to inform future fire-risk mitigation.

II. Materials and Methods

A. Data

Soil moisture observations were obtained from the Peatland Restoration Information and Monitoring System (PRIMS) maintained by the Peatland and Mangrove Restoration Agency (BRGM). The 2023 dataset contains volumetric soil moisture measurements collected at monitoring stations across Indonesian peatlands [19]. The analysis focused on Riau, Jambi, and South Sumatra, where Sumatra's peatlands are concentrated.

Data preprocessing was carried out to ensure spatial consistency. Starting from 8,966 raw records, 97 measurements outside the physically valid range (0–100%) were removed, and records sharing identical coordinates were averaged, yielding 7,584 unique spatial points. Geographic coordinates were converted into point geometries using GeoPandas, and each point was assigned to its province and regency/city through a spatial join with administrative boundaries.

To restrict the analysis to genuine peatland, only points falling inside mapped peat polygons were retained through a spatial join with the Indonesian peatland map. In addition, one Riau peat polygon that consisted of a cluster of repeated, gap-filled values was excluded, because such imputed records do not reflect true spatial variability and would distort variogram estimation. This peatland masking rather than any random subsampling accounts for the reduction in the number of points per province. The final numbers of observations used for interpolation were 490 for Riau, 319 for Jambi, and 3,322 for South Sumatra.

It should be noted that in Jambi a single repeated value (55.2%) accounts for about 87% of the retained points, a residual of the imputation present in the source data. This limits the true spatial variability in Jambi and is considered when interpreting the Jambi results.

B. Experimental Semivariogram

The experimental semivariogram illustrates the relationship between semivariance $\gamma(h)$ and lag distance h. The experimental semivariogram is constructed from observed data to quantify spatial dependence as the first step of geostatistical modelling [20]. Semivariance is initially defined as:

$$\gamma(x_i - x_j) = \frac{1}{2}\, var(Z_{x_i} - Z_{x_j}) \tag{1}$$

where *var* denotes the variance of the difference between measurements at points i and j. In practice, the semivariogram is calculated with the empirical estimator [20]:

$$\gamma(h) = \frac{1}{2N_h}\sum_{i=1}^{N_h}[Z(x_i) - Z(x_i + h)]^2 \tag{2}$$

where N_h represents the number of data pairs separated by the lag h. The plotted semivariance values for each lag form the experimental semivariogram, which is used to analyze the spatial structure and continuity of the dataset.

C. Semivariogram Fitting

The experimental semivariogram describes how semivariance changes with lag distance, but it is defined only at discrete lags and is not guaranteed to form a valid spatial model. It therefore cannot be used directly for prediction. To obtain a continuous and valid model, a theoretical variogram with a defined mathematical form is fitted to the experimental values [21], [22]. The most commonly used theoretical models include the spherical, exponential, and Gaussian forms [22].

In this study, all three models were fitted to the experimental semivariogram of each province, and their parameters (nugget, sill, and range) are reported in Table II. For each province a single model was adopted for interpolation, guided by the fit to the experimental semivariogram and by the cross-validation performance: the spherical model for Riau and Jambi, and the exponential model for South Sumatra.

D. Ordinary Kriging

Ordinary kriging estimates soil moisture at unsampled locations at an unknown location by weighting nearby measured points, assuming a constant but unknown mean across the study area[23]. It is formulated as:

$$Z(x_0) = \sum_{i=1}^{n} \lambda_i Z(x_i) \tag{3}$$

where $\sum\lambda_i = 1$. Here, $Z(x_0)$ represents the estimated value at the target location x_0, $Z(x_i)$ denotes the observed value at point i, and λ_i are weights derived from the variogram model.

E. Universal Kriging

Universal kriging (UK) is applied when the data exhibit non-stationarity in the form of a large-scale spatial trend, in contrast to the constant-mean assumption of ordinary kriging. The variable is modeled as the sum of a deterministic trend and a spatially correlated residual:

$$\mathrm{Z(s)} = \mu(s) + \varepsilon(s) \tag{4}$$

where $\mu(s)$ is expressed as a linear combination of coordinate-based basis functions $P_k(s)$ with coefficients b_k [24] :

$$\mu(s) = \sum_{k=1}^{l} b_k P_k(s) \quad (5)$$

and the residual $\varepsilon(s)$ is assumed to be intrinsically stationary, so that its spatial dependence can be described by a single variogram. In this study the trend order was set per province according to the spatial behavior observed in each dataset: a first-order (linear) trend in the coordinates for Riau, and a second-order (linear plus quadratic) trend for Jambi and South Sumatra. Ordinary kriging is recovered as the special case in which the trend reduces to an unknown constant mean.

F. Evaluation Metrics

Model performance was evaluated using spatial-block cross-validation. Because soil moisture is spatially autocorrelated, a random division of points would place neighboring observations in both the training and validation sets, leading to information leakage and over-optimistic error estimates. To avoid this, the study area of each province was divided into square blocks of 20 km, and whole blocks were assigned to one of five folds. In this way, all points within a block were held out together, so that validation always measured prediction at locations spatially separated from the training data.

Prediction performance was assessed using the mean absolute error (MAE), root mean square error (RMSE), and mean standardized squared deviation (MSDR). Lower RMSE and MAE indicate higher accuracy; and MSDR values close to one indicate well-calibrated kriging uncertainty, with values above one indicating under-estimated uncertainty and values below one indicating over-estimated uncertainty [21], [25]. The metrics are defined as:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|z(x_i) - \hat{z}(x_i)| \quad (6)$$

$$RMSE = \left[\frac{1}{n}\sum_{i=1}^{n}(z(x_i) - \hat{z}(x_i))^2\right]^{1/2} \quad (7)$$

$$MSDR = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{z(x_i) - \hat{z}(x_i)}{\sigma(x_i)}\right)^2 \quad (8)$$

where $z(x_i)$ denotes the observed soil moisture, $\hat{z}(x_i)$ denotes the predicted soil moisture, and $\sigma(x_i)$ represents the kriging standard deviation associated with location.

G. Visualization and Interpretation

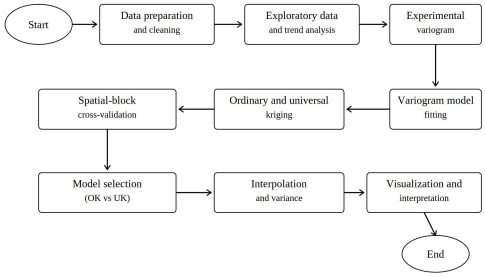


Fig. 1. Flowchart of Kriging Algorithm Implementation

The flowchart in Fig. 1 summarizes the geostatistical workflow adopted in this study. The process begins with data preparation and preprocessing, followed by exploratory data analysis and experimental semivariogram construction. A theoretical variogram model is then fitted to characterize spatial dependence. Using the selected variogram model, ordinary kriging and universal kriging are performed and evaluated through spatial cross-validation using RMSE, MAE, and MSDR. Finally, the selected interpolation results are visualized and interpreted to analyze the spatial distribution of peatland soil moisture. All analyses were performed in Python. Variogram fitting and both ordinary and universal kriging used the PyKrige library; GeoPandas was used for spatial preprocessing, and scikit-learn for the cross-validation procedure and the trend computation.

III. Results and Discussions

A. Sampling Points

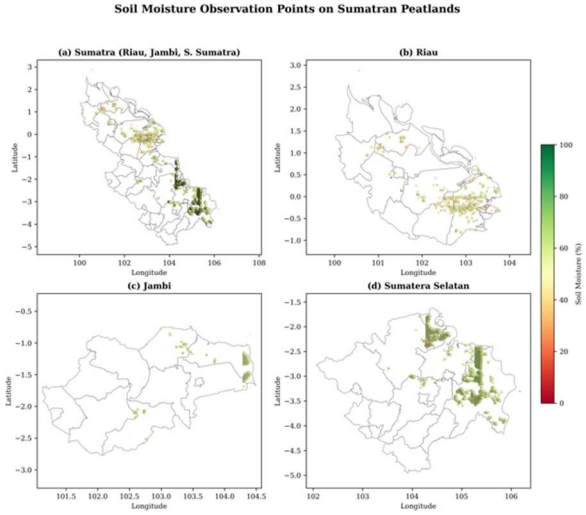


Fig. 2. Soil moisture observation points on Sumatran peatlands: (a) all three provinces, (b) Riau, (c) Jambi, (d) South Sumatra

Figure 2 shows the spatial distribution of monitoring stations in the three study provinces. South Sumatra contains by far the largest number of observations (3,322), followed by Riau (490) and Jambi (319). This uneven distribution, together with the clustered arrangement of stations within each province, influences the reliability of variogram estimation and interpolation accuracy, and is considered when interpreting the provincial results.

B. Descriptive Statistics

Table I summarizes the descriptive statistics of the cleaned dataset. The mean soil moisture is 54.06% with a standard deviation of 5.54%. The median value (55.17%) coincides with a repeated imputed value observed during preprocessing; its dominance in the distribution should therefore be interpreted with caution rather than as evidence of a naturally near-normal field. Recorded values range from 31.20% to 76.69%, with the lower part of this range indicating peatland areas in markedly drier conditions. The coordinate statistics (longitude and latitude) are included to describe the spatial extent of the sampling domain across the three provinces.

TABLE I. DESCRIPTIVE STATISTICS OF SOIL MOISTURE AND GEOGRAPHICAL COORDINATES PER PROVINCE: RIAU 490, JAMBI 319, SOUTH SUMATRA 3,322 SPOTS

	Soil Moisture (%)	Longitude	Latitude
Count	4131	4131	4131
Mean	54.058	107.206	-2.422
Std	5.544	3.79	1.048
Min	31.2	100.734	-3.992
25%	51.3	104.418	-3.109
50%	55.17	105.328	-2.685
75%	56.3	111.86	-2.135
Max	76.689	115.401	1.515

C. *Semivariogram Analysis*

The experimental semivariograms of the three study provinces were analyzed to characterize spatial dependence and identify the most appropriate theoretical variogram model for kriging interpolation. Spherical, exponential, and Gaussian models were fitted to each experimental semivariogram, and their parameters are reported in Table II. For each province a single model was adopted for interpolation, guided by both the fit to the experimental semivariogram and the cross-validation performance.

TABLE II. FITTED VARIOGRAM PARAMETERS

Province	Model	Nugget	Sill	Range (km)
Riau	Spherical	37.31	53.81	145.5
Riau	Exponential	36.86	66.71	145.5
Riau	Gaussian	39.39	52.18	62.7
Jambi	Spherical	0.00	6.97	145.5
Jambi	Exponential	0.00	11.35	145.5
Jambi	Gaussian	0.00	38.15	145.5
S.Sumatra	Spherical	2.89	8.60	39.6
S.Sumatra	Exponential	1.74	8.70	13.7
S.Sumatra	Gaussian	3.73	8.62	19.6

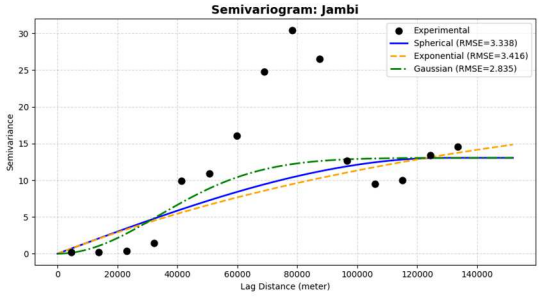


Fig. 3. Semivariogram analysis plot in Jambi

As illustrated in Fig. 3, the empirical semivariogram of Jambi reveals a distinct spatial dependence that is well represented by the spherical model. Compared with the exponential and Gaussian alternatives, the spherical model provided the most reliable representation of the observed pattern and was therefore adopted for subsequent interpolation. It should be noted, however, that the very clean semivariogram of Jambi, with a near-zero nugget, partly reflects the low variability of the data, which are dominated by a single repeated value rather than a naturally smooth field.

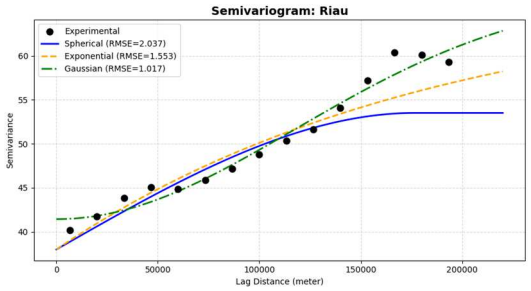


Fig. 4. Plot semivariogram analysis in Riau

Unlike Jambi, the semivariogram obtained for Riau (Fig. 4) exhibits a more gradual increase in semivariance with lag distance, suggesting weaker spatial continuity among observations. Despite the similar performance of the candidate models, the spherical model achieved the best validation results and was selected for kriging. This weaker spatial dependence is confirmed by the fitted parameters in Table II: the nugget (≈ 37) is close to the sill (≈ 54), meaning that most of the variance is already present at short lag distances and little exploitable spatial structure remains. This near-nugget behavior explains the comparatively high prediction error obtained for Riau in the validation stage.

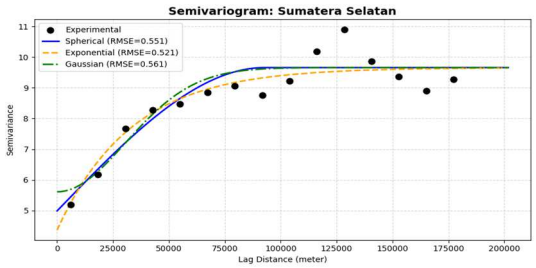


Fig. 5. Semivariogram analysis plot in South Sumatra

A different pattern is observed in South Sumatra (Fig. 5), where the empirical semivariogram is better represented by the exponential model. The fitted curve follows the experimental semivariogram more closely than the spherical and Gaussian models, indicating that the exponential model is more appropriate for describing the spatial variability in this province. This finding is further supported by the cross-validation results, which indicate comparable RMSE, MAE, and MSDR values for ordinary and universal kriging.

The possibility of spatial anisotropy was examined using directional experimental semivariograms in four directions (0°, 45°, 90°, and 135°). In Riau, the directional sills were similar (ratio ≈ 1.25), so an isotropic model is reasonable. In South Sumatra, a mild anisotropy was observed (ratio ≈ 1.6), with slightly longer spatial continuity in one direction; this was retained as a limitation rather than modeled explicitly, since the effect was moderate. In Jambi, the directional analysis was inconclusive, as several directions contained too

few point pairs to be reliable and the dominance of a single repeated value produced an unstable ratio. Overall, the isotropic assumption is adequate for Riau, acceptable for South Sumatra, and unresolved for Jambi given the data limitations.

D. Kriging Interpolation Mapping

The variogram models selected from the semivariogram analysis were subsequently used to interpolate peatland soil moisture using ordinary kriging (OK) and universal kriging (UK). Figure 6 juxtaposes the prediction surfaces generated by both methods for Jambi, Riau, and South Sumatra, allowing their spatial patterns to be compared under identical interpolation settings.

At first glance, the interpolation surfaces produced by ordinary and universal kriging are remarkably alike across all three provinces. Large-scale moisture patterns, local gradients, and transition zones remain nearly unchanged regardless of whether a deterministic drift is incorporated into the kriging formulation. The close visual agreement mirrors the cross-validation results, where both methods produced similar spatial patterns with province-dependent differences in predictive performance. This suggests that the representation of spatial dependence through the variogram is an important factor in this dataset, while the benefit of deterministic drift remains region-specific.

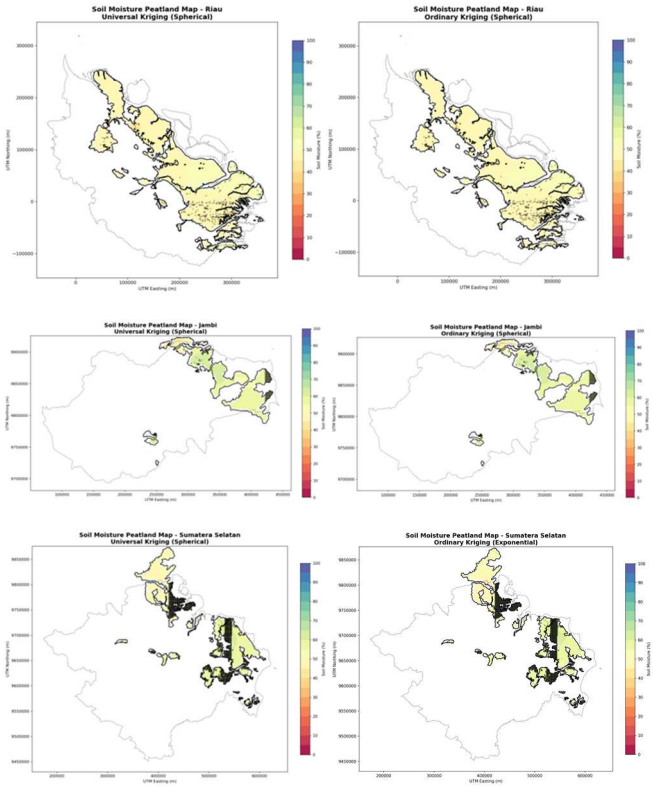


Fig. 6. Ordinary versus universal kriging soil moisture maps for Riau, Jambi, and South Sumatra (left: UK; right: OK)

The Riau maps are characterized by broad regions of relatively uniform moisture with only subtle local variations. This smoothness reflects a property of the data rather than a limitation of the method: because the nugget is close to the sill, little exploitable spatial structure remains, and the kriging predictor tends toward the global mean. This explains both the high prediction error and the uniform appearance of the surface. Introducing a deterministic trend produces little visible change, and both kriging approaches recover nearly identical distributions.

A different behavior emerges in Jambi, where the predicted surface is dominated by relatively homogeneous moisture values over much of the peatland extent. Rather than indicating poor interpolation performance, this pattern reflects the limited variability preserved after data preprocessing. Because neighboring observations share similar moisture levels, both ordinary and universal kriging reconstruct comparable spatial fields, with only minor local differences between the two methods. This limited variability is itself a consequence of the imputation in the source data, where a single repeated value dominates the Jambi observations; the smooth Jambi surface should therefore be read with this caveat in mind.

In contrast, South Sumatra exhibits a richer spatial texture, with distinct moisture gradients extending across the peatland area. These gradients are reproduced consistently by both kriging methods, indicating that the exponential variogram successfully captures the underlying spatial dependence. The visual similarity between the two prediction surfaces further supports the numerical evidence obtained from cross-validation, where only marginal differences were observed between OK and UK.

To assess whether environmental variables could explain the moisture patterns beyond location alone, the relationship between soil moisture and several covariates elevation, slope, a topographic wetness index, and distance to the nearest river was examined. The correlations were generally weak across all provinces. The only moderate association was a negative correlation between soil moisture and distance to rivers in Jambi ($r \approx -0.42$), consistent with the drying influence of drainage away from watercourses. These generally weak relationships indicate that soil-moisture variation in these peatlands is not strongly controlled by topography at the scale of the available data, and they help explain why incorporating a deterministic trend whether based on coordinates (UK) or on covariates offers little improvement over ordinary kriging.

Taken together, the interpolation maps demonstrate that the primary determinant of prediction quality is not the choice between ordinary and universal kriging, but the spatial dependence represented by the fitted variogram model. Once this dependence is adequately characterized, incorporating a deterministic drift contributes only modest refinements to the interpolated moisture field. This observation is consistent across the three study provinces and reinforces the quantitative findings obtained through cross-validation.

E. Cross-Validation

While the interpolation maps suggest only minor visual differences between ordinary and universal kriging, quantitative evaluation is required to confirm this. Table III

reports the spatial-block cross-validation results for both methods.

TABLE III. SPATIAL-BLOCK CROSS-VALIDATION RESULTS FOR ORDINARY AND UNIVERSAL KRIGING

Province	Variogram	Model	RMSE	MAE	MSDR
Riau	Spherical	OK	5.79	5.21	1.14
Riau	Spherical	UK	6.80	5.45	1.14
Jambi	Spherical	OK	1.20	0.39	0.42
Jambi	Spherical	UK	1.01	0.22	0.83
S.Sumatra	Exponential	OK	1.38	1.21	1.46
S.Sumatra	Exponential	UK	2.49	1.61	2.57

Across the three provinces, the relative performance of ordinary and universal kriging varied by region. In Riau, OK achieved lower prediction errors than UK (RMSE: 5.79 vs. 6.80), whereas UK improved accuracy in Jambi (RMSE: 1.20 vs. 1.01). South Sumatra showed the opposite pattern, where OK produced lower errors than UK (RMSE: 1.38 vs. 2.49). These differences indicate that incorporating a deterministic drift term does not consistently improve prediction performance and depends on the spatial characteristics of each province.

The provinces also differed markedly in overall predictability. Jambi exhibited the lowest prediction errors, followed by South Sumatra, while Riau showed the highest error. These differences are consistent with the spatial dependence identified from the variogram analysis, particularly the near-nugget behavior observed in Riau, which indicates weak spatial autocorrelation rather than a limitation of the kriging approach itself.

Uncertainty calibration also varied among provinces. Riau showed MSDR values close to one, indicating well-calibrated uncertainty estimates. Jambi produced lower MSDR values under OK (0.42), while UK improved the calibration (0.83), approaching the ideal value of one. In contrast, South Sumatra showed higher MSDR values, particularly under UK (2.57), suggesting underestimated uncertainty when predicting spatially separated blocks. These results highlight that uncertainty calibration and spatial dependence characterization are important considerations alongside the choice between OK and UK. However, these findings are limited to the three investigated provinces and should not be generalized to other peatland regions.

F. Future Research

Future research can extend this work in several complementary directions. First, space–time kriging [26], [27], [28], [29], [30] could capture the seasonal and inter-annual dynamics of soil moisture that a single-year snapshot cannot represent. Second, denser and more evenly distributed sampling, particularly in Riau where weak spatial structure limited predictability, and improved handling of incomplete observations in Jambi, together with the incorporation of environmental covariates such as land use, drainage, peat depth, and rainfall, would strengthen variogram estimation, uncertainty characterization, and interpolation performance. Furthermore, integrating geostatistical outputs with additional analytical frameworks could extend the application of soil-moisture maps beyond spatial interpolation. For example, Correspondence Analysis [31], [32] may be explored to investigate associations between categorical peatland characteristics, such as land-use type, drainage intensity, and restoration status, and discretized soil-moisture conditions. These relationships may support the development of future fire-risk assessment frameworks by identifying environmental conditions associated with critical moisture regimes [33], [34], [35]. In addition, economic evaluation approaches [36], [37] may be considered to assess the long-term efficiency of peatland monitoring and restoration strategies. These extensions would help connect the geostatistical maps developed in this study with broader peatland management and decision-support applications.

IV. CONCLUSION

This study compared ordinary kriging (OK) and universal kriging (UK) for peatland soil moisture interpolation in three major peatland provinces of Sumatra using the 2023 PRIMS dataset. Province-specific semivariogram models were selected through variogram fitting and applied to generate spatial interpolation maps, and predictive performance was evaluated using spatial-block cross-validation based on RMSE, MAE, and MSDR. Both methods produced comparable spatial patterns across the study areas; however, their predictive performance varied among provinces. UK improved prediction accuracy in Jambi, whereas OK achieved lower errors in Riau and South Sumatra. These results indicate that incorporating a deterministic drift does not consistently improve peatland soil moisture interpolation, and its effectiveness depends on the spatial characteristics of each region. Instead, accurate representation of spatial dependence through variogram modeling and appropriate validation strategies remains essential for reliable interpolation performance.

Overall, this study provides a systematic geostatistical assessment of peatland soil moisture interpolation in Sumatra and demonstrates the applicability of OK and UK for province-scale mapping at a 5 km resolution within the mapped peatland domains of each province. The resulting maps provide spatially continuous information that can support peatland monitoring and serve as an input to future fire-risk assessment, rather than constituting a direct fire-risk assessment themselves. These findings are based on a single-year (2023) dataset and three provinces with uneven sampling characteristics and differences in data quality among regions; therefore, the results should be interpreted as province- and season-specific representations rather than a general long-term peatland moisture regime.

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